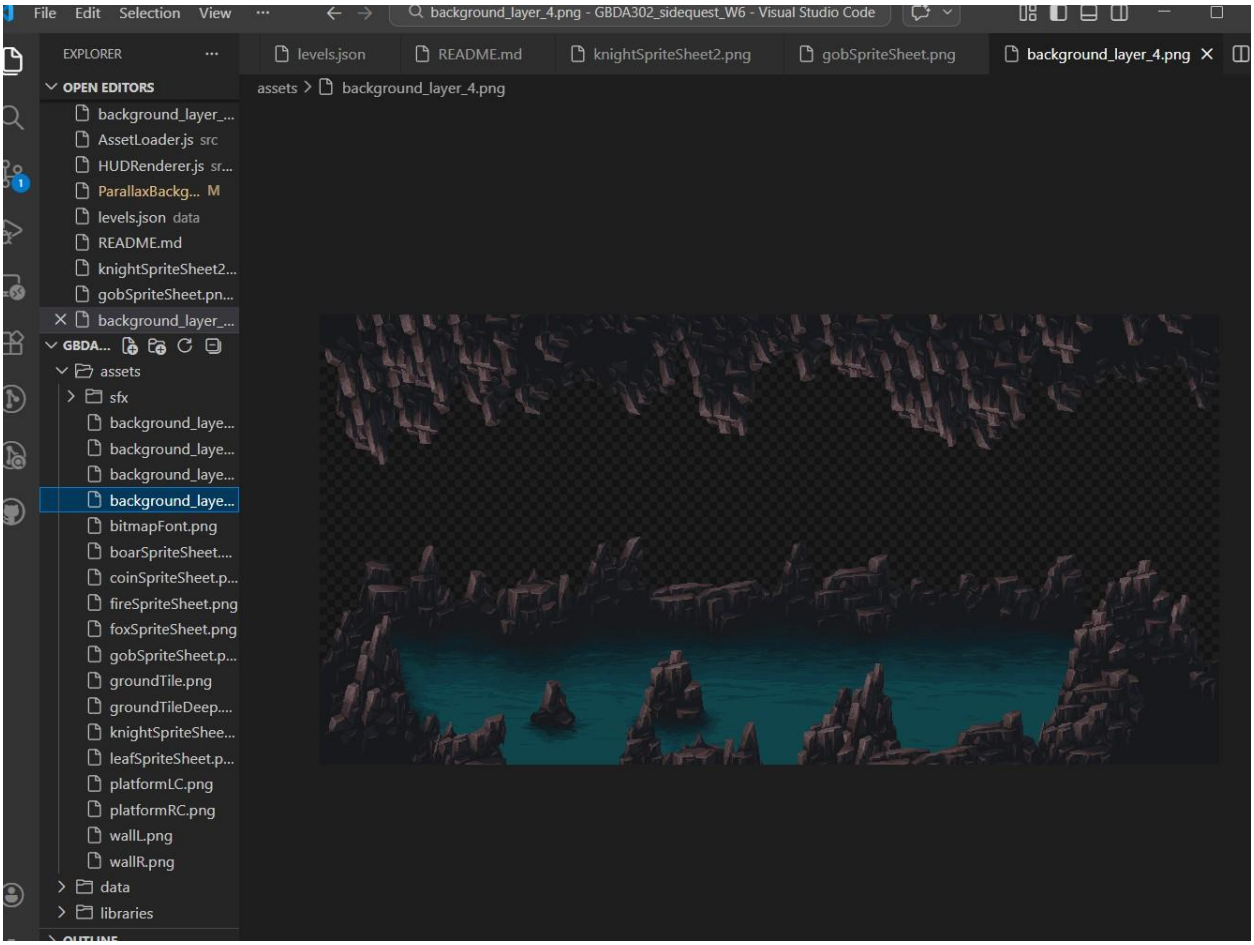
Process s Decision Documentation

Process Overview Visualization (Group Work Only)



Project/Assignment Decisions

I chose to change these levels' sprite packs to make it look more adventurous and like the player was exploring a cavern in order to retrieve coins.

Side Quests and A4 (Individual Work)

- I changed the player sprite to a slime and changed the setting to be a cave.
- I did to give the feeling of adventure.
- For the audio, I chose to use soundbites from Link to add more 'character' to the world

GenAI Documentation

Everyone must complete this section. If not GenAI was used, write, "No GenAI used for this task." When GenAI is not used, process evidence should still demonstrate iteration, revision, or development over time.

Because GenAI can closely mimic human-created work, instructors or TAs may occasionally request additional process evidence to confirm non-use. This may include original working files (e.g., an illustrator file), intermediate drafts, or a brief check-in with a TA to walk through your process.

These requests are not an assumption of misconduct. They are part of ensuring academic integrity in an environment where distinguishing between human-created and AI-generated work is increasingly difficult.

If GenAI was used (keep each response as brief as possible):

Date Used: 11th of March, 2026

Tool Disclosure: Claude Sonnet 4.6

Purpose of Use: I used GenAI to fix the issues in my parallax system.

Summary of Interaction: I used the tool to check my parallax system because it didn't move the way I needed it to. It then fixed what I was missing from the files and gave me the way to fix it.

Human Decision Point(s): I had to make some adjustments in the code and also ask for clarification since I gave it the original example 5 file and not the one that I had changed to have 4 background layers.

Integrity & Verification Note: I checked the integrity and validity of the code by making sure that it only used the information from the files I provided.

Scope of GenAI Use: The tool was only used for the background and parallax building and tweaking.

Limitations or Misfires: After inputting what the GenAI tool gave me, I noticed that it duplicated one of my sprites and was not able to change it back.

Summary of Process (Human + Tool)

- Iteration between drafts or versions
- Discussion with my classmates, and asking them for help with some code fixes
- Testing and revision
- Moments of uncertainty, failure, or rework: I had to spend a lot of time fixing, combining and rescaling the character sheets because all the different actions were split up into separate PNGs.

Decision Points & Trade-offs

Describe one or two key decisions you made:

- What options you considered
- What changed
- Why that choice was made

These decisions should align with decision points shown in your visualization (for A1 – A3).

Verification & Judgement

- Playtesting
- Peer feedback
- Comparison with course examples
- Re-reading assignment criteria

Limitations, Dead Ends, or Open Questions

Note anything that:

- The character sheets needed to be the same size as what was decided in the assets, so I had to find one that wasn't too small, or that wouldn't crop weirdly.

Appendix

Prompt:

"Can you help me fix my code since i want my parallax system to work. ParallaxBackground.js, main.js"

Response:

Great, I can see your full codebase clearly! This is a well-architected p5play project. Let me check a couple more files to get the full picture before touching anything.

See all uploaded files
See all uploaded files

To give you the most accurate help, it would be really useful to see a few more files:

1. **`data/levels.json`** — this is critical for the parallax fix. Your `ParallaxBackground` reads `level.view.parallax` from it, and I need to see how your layers are currently defined there (the `img`, `speed`, and `key` fields).

That said, here's what I can already tell you about both issues:

Parallax — The system looks correct in ParallaxBackground.js, but the most common reason it breaks is a mismatch in levels.json. The img field in your parallax layer definitions needs to be a path relative to your HTML file (e.g. "assets/background_layer_1.png"), and the layers need a speed (not factor) field since main.js reads d.speed.

Promp2:

“levels.json, tuning.json”

Response:

Perfect, I have everything I need. Let me analyze the issues:

1. **Parallax**: Your levels.json has 3 layers but you have 4 background images. Also, LevelLoader.js reads doc.viewTilesW/H but your levels.json stores dimensions inside level.view as w/h — so viewW/viewH may be computing wrong, causing parallax to draw at the wrong size.

Let me fix both:

Check levels.json structure for viewW/H computation

Check levels.json structure for viewW/H computation

Confirmed — doc.viewTilesW is null, so LevelLoader computes viewW = 24 * 10 = 240 and viewH = 24 * 8 = 192 from defaults, which actually matches. The parallax issue is that you only have 3 layers in levels.json but 4 background images, and the ParallaxBackground tiles images using img.width but the image height isn't being scaled to viewH properly. Let me trace the full draw call and fix everything now.

Copy files to work on them

Copy files to work on them

Here are the exact changes needed. I'll give you **2 targeted fixes**:

Fix 1 — levels.json: Add your 4th background layer

Add 4th parallax layer to match the 4 background images

Add 4th parallax layer to match the 4 background images

Fix 2 — ParallaxBackground.js: The draw loop tiles by img.width but images may not be loaded yet when .width is read. Also the image is drawn with viewH for height which is correct, but the tiling needs a left-side buffer to avoid a gap when scrolling left.

Now copy all edited files to outputs:

Ran a command, read a file
Ran a command, read a file

Here's a summary of every change and where to place each file:

`levels.json` — Added the 4th parallax layer (`background_layer_4.png`) with a `speed` of `0.6`, and adjusted the other speeds to `0.1/0.2/0.4/0.6` for a more natural depth feel.

`ParallaxBackground.js` — Fixed the tiling math. The old code could produce gaps when scrolling left. The new version uses modulo to keep the offset in a clean `[0, imgW)` range and always starts one tile to the left of the viewport edge.

Levels

Code · JSON

Parallaxbackground

JS

Main

JS

Prompt3:

for this part in Main.js "`// Background colour is per-level in levels.json: level.view.background const bg = levelPkg.level?.view?.background ?? [69, 61, 79]; background(bg[0], bg[1], bg[2]);`" aren't there meant to be 4 background layers

Response:

Good catch, but actually that `background()` call is fine as-is — it's not a parallax layer, it's just the solid **canvas fill colour** drawn before anything else (the dark purple `[69, 61, 79]`). It acts as a fallback in case the parallax images don't fully cover the screen.

The 4 parallax image layers are handled separately right after it:

```
background(bg[0], bg[1], bg[2]);  // ← solid colour fill (1 call)

parallax?.draw({ ... });           // ← draws all 4 image layers on top
```

So the flow each frame is:

1. Fill canvas with solid purple
2. Draw `background_layer_1.png` (slowest, farthest back)
3. Draw `background_layer_2.png`
4. Draw `background_layer_3.png`
5. Draw `background_layer_4.png` (fastest, closest)
6. Draw world sprites on top of all that

Your `levels.json` now has all 4 layers listed, and `ParallaxBackground.js` loops through all of them — so nothing is missing there. The solid colour is just insurance. You're good!

Also — good news on the jump sound: your `PlayerController.js` already has `this.events?.emit("player:jumped", {})` on line 63, so that event is already being emitted. The `Game.js` fix I made will catch it perfectly. No changes needed to `PlayerController.js`.